If one asks whether 2 is a complex number, many students hesitate to say "yes". As every integer is a rational number, we know that every real number is also a complex number. So to clearly specify a complex number that is not a real number, that is to specify numbers of form $\alpha + i\beta$ with $\beta \neq 0$, we use the term **"non-real complex number".** Some authors call such a number an **imaginary number**.

Remark 1

Let $z_0 = \alpha + i\beta$ with $\beta \neq 0$. Then $\bar{z}_0 = \alpha - i\beta$. If $\alpha + i\beta$ is a root of a polynomial equation $P(x) = 0$ with real coefficients, then by Complex Conjugate Root Theorem, $\alpha - i\beta$ is also a root of $P(x) = 0$. Usually the above statement will be stated as **complex roots occur in pairs;** but actually it means that **non-real complex roots or imaginary roots occur as conjugate pairs, being the coefficients of the polynomial equation are real.**

Remark 2

From this we see that any odd degree polynomial equation with real coefficients has at least one real root; in fact, the number of real roots of an odd degree polynomial equation with real coefficients is always an odd number. Similarly the number of real roots of an even degree polynomial equation with real coefficients is always an even number.

Example 3.8

Find the monic polynomial equation of minimum degree with real coefficients having $2 - \sqrt{3}\,i$ as a root.

Solution

Since $2 - \sqrt{3}\,i$ is a root of the required polynomial equation with real coefficients, $2 + \sqrt{3}\,i$ is also a root. Hence the sum of the roots is 4 and the product of the roots is 7. Thus $x^2 - 4x + 7 = 0$ is the required monic polynomial equation. ■

3.4.2 Irrational Roots

If we further restrict the coefficients of the quadratic equation $ax^2 + bx + c = 0$ to be rational, we get some interesting results. Let us consider a quadratic equation $ax^2 + bx + c = 0$ with a, b, and c rational. As usual let $\Delta = b^2 - 4ac$ and let r_1 and r_2 be the roots. In this case, when $\Delta = 0$, we have $r_1 = r_2$; this root is not only real, it is in fact a rational number.

When Δ is positive, then no doubt that $\sqrt{\Delta}$ exists in $\mathbb{R}$ and we get two distinct real roots. But $\sqrt{\Delta}$ will be a rational number for certain values of a, b, and c, and it is an irrational number for other values of a, b, and c.

If $\sqrt{\Delta}$ is rational, then both r_1 and r_2 are rational.

If $\sqrt{\Delta}$ is irrational, then both r_1 and r_2 are irrational.

Immediately we have a question. If $\Delta > 0$, when will $\sqrt{\Delta}$ be rational and when will it be irrational? To answer this question, first we observe that Δ is rational, as the coefficients are rational numbers. So $\Delta = \frac{m}{n}$ for some positive integers m and n with $(m, n) = 1$ where (m, n) denotes the

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